

Mukesh Kumar

Software Engineer

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I am a qualified Software Engineer from India with over 2+ years of experience. I have hands-on experience in Python frameworks like Django and Flask, Microsoft SQL, and ML/AI, including Generative AI. With strong critical thinking skills and a passion for learning, I contribute effectively to organizational goals. I also have experience working in cross-cultural environments and successfully coordinating with teams in the USA.

Education

- **Master in Computer Applications**
- **Bachelor in Computer Applications**

Experience

Software Engineer

Signity Software Solutions Pvt. Ltd.

Jan 2025 - Current , Mohali Punjab

AI/ML Engineer with experience in building scalable, production-ready AI systems using NLP, computer vision, and multimodal models. Specialized in model fine-tuning, inference optimization, and retrieval-based pipelines. Developed high-performance FastAPI backends with secure authentication, async processing, and efficient data handling. Designed end-to-end AI workflows including document processing, semantic search, and LLM-powered responses. Collaborated cross-functionally to deliver enterprise-grade solutions.

Jr. Software Engineer

C.S. Soft Solutions (India) Private Limited

July 2023 -December 2024 ,Mohali Punjab

Developed and deployed custom web applications using Python, AI/ML, and .NET technologies, optimizing user interactions and increasing engagement by 25%. Enhanced system performance by streamlining API interactions, resulting in improved overall efficiency and smoother user experience. Completed industrial training at CS Soft Solutions, where I worked on test tasks and assignments, learned project-based technologies, and gained knowledge transfer (KT) on existing projects like MOA, contributing to project testing and development.

Internship

CS Infotech

December 2022 - June 2023

Sector 22 C, Chandigarh

Developed an eCommerce web application as an intern using ASP.NET Core MVC, implementing authentication with Identity Framework, OAUTH, and integrating Stripe for secure payment transactions. Contributed to the development of a Web API in ASP.NET Core, enhancing communication between frontend and backend applications and improving integration and functionality.

Profile Summary

- Proficient in languages such as Python, JavaScript, C# and TypeScript .
- Experienced in frameworks and libraries like Flask, Django, .Net and Fastapi.
- Skilled in working with databases including PostgreSQL, MySQL, VectorDB, MongoDB, SQL Alchemy and SQLite.
- Knowledgeable in web technologies such as HTML, CSS, XML, JSX, RESTful APIs, and GraphQL.
- Familiar with tools like Git, Postman, Socket.io.
- Successfully designed and developed software solutions at Signity Software Solutions Mohali, consistently meeting client requirements and ensuring functionality, scalability, and performance.
- Demonstrated expertise in Python leading the development of user-friendly interfaces and backend functionalities for various applications.
- Integrated third-party tools and components to enhance the functionality of end-user applications across different platforms.
- Led the development of Dashworks Search, a MacOS application connecting various third-party platforms using technologies like Python, React.
- Developed Activue App, a multi-platform entertainment app for elderly residents, utilizing React Native, Python, and fast.
- Created a web-based admin panel (Activue Admin Panel) for seamless control and monitoring of the Activue App using React.js and Python.

Technical Skills

- **Language:** Python, JavaScript, TypeScript, C#,.
- **Frameworks/Libraries:** Django , Flask , REST API , Dot Net , MVC , Razor Pages
- **Databases:** PostgreSQL, MySQL , MongoDB, SQLite,QDRANTDB,FAISS.
- **Web Technologies:** HTML, CSS, XML, JSX, RESTful APIs
- **Tools:** Git, Postman, Socket.io.

1. Full-Stack Development:

- Develop and maintain scalable web applications using **Python (Flask/Django), ReactJS, and .NET Core**.
- Build and integrate RESTful APIs to ensure seamless communication between frontend and backend.
- Design and optimize database interactions for better performance and maintainability.
- Develop interactive dashboards and user management features for applications like **Solstice Health, SNiP Nutrigenomics, and MOA WebApp**.
- Implement role-based access control and authentication for admin modules and user subscriptions.

2. Backend Development & API Integration:

- Develop backend systems using **Python, Flask, Django, and C# (.NET Core)**.
- Optimize data flow and API performance for AI-driven platforms such as **Solstice Health and SNiP Nutrigenomics**.
- Work on **CRUD operations, database management, and optimization** in platforms like **Mill Data Admin & Forisk's Mill Capacity Database**.
- Create and optimize **stored procedures and SQL queries** to enhance database efficiency.
- Integrate external APIs such as **Saavn API for Music Recommendation System** and third-party DNA report processing for **SNiP Nutrigenomics**.

3. Machine Learning & AI Development:

- Implement **AI-driven recommendation systems**, such as the **Music & Movie Recommendation System** using **NLP and Cosine Similarity**.
- Develop **ML models for predictive analytics**, including the **Cuisine Predictor using Logistic Regression**.
- Work on **GPT-4 & LLM-based AI solutions** for **Solstice Health**, enhancing **content evaluation and generation**.
- Build **computer vision and NLP-based solutions** to analyze user data and provide intelligent recommendations.

4. Performance Optimization & Scalability:

- Optimize database queries and **reduce query load by 30%** using **stored procedures and indexing**.
- Enhance **web application performance by 20%** through **ReactJS, jQuery, and AJAX optimizations**.
- Develop **views and backend modules** to streamline business workflows and reduce operational overhead.

Projects

1. Envisify (BERT Base Classification)

Role: ML Engineer

Description: Envisify is an advanced text analysis application designed to enhance enterprise capabilities by fine-tuning on diverse text datasets. This tool leverages BERT Base Classification to analyze ethnicity, age groups, and sentiment. Additionally, the application incorporates GPT prompt functionality and a Next.js frontend for dynamic and efficient text analysis.

Roles and Responsibilities:

- Designed and implemented text analysis functionalities using Python and Next.js.
 - Fine-tuned BERT models to analyze ethnicity, age groups, and sentiment.
 - Collaborated on the integration of various third-party platforms.
 - Ensured seamless connectivity and efficient text analysis processes.
 - Incorporated GPT prompt functionality and developed the frontend using Next.js.
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2. Plan and Proof (Fine-tuning the Qwen2.5-VL Model Series)

Role: AI Engineer

Description: Plan and Proof is an AI-powered system designed to analyze building plans and architectural drawings to determine whether they comply with the **National Construction Code (NCC) of Australia**. It functions like an automated building inspector capable of reading PDF drawings, extracting technical details, and identifying compliance issues based on Australian building standards and regulations.

Roles and Responsibilities:

- **Developed and maintained a FastAPI backend**, serving as the main API layer for processing PDF drawings, handling file uploads, and exposing model inference endpoints.
- **Integrated Hugging Face Transformers**, enabling loading, configuring, and managing the Qwen2.5-VL model series for multimodal (text + vision) analysis.
- **Used PyTorch for model inference and fine-tuning**, including training LoRA adapters, optimizing model performance, and running GPU-accelerated computations.
- **Applied Pillow (PIL) for image preprocessing**, such as resizing, cropping, and converting plan drawings into model-ready formats.
- **Utilized pdf2image to convert PDF plan pages into high-resolution images**, enabling the AI model to interpret architectural drawings effectively.

- **Implemented PyMuPDF (fitz) to extract metadata, text blocks, and structural information** from PDF files during pre-processing and validation.
 - **Used python-multipart for reliable handling of large PDF uploads** through FastAPI endpoints.
 - **Implemented aiofiles to perform asynchronous file operations**, improving performance during large file reads/writes and concurrent uploads.
 - **Used json-repair to parse and fix invalid or incomplete JSON outputs** generated by the LLM, ensuring stable downstream processing.
 - **Applied PEFT with LoRA for efficient fine-tuning** of Qwen2.5-VL models, enabling custom training on architectural drawings and NCC compliance datasets without requiring full-model training.
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3. Wisdom-AI (RAG-Based System)

Role: AI Engineer

Description:

WisdomAI is a RAG-powered business intelligence platform that enables users to interact with their data using natural language. It works like a personal data analyst—allowing users to ask questions, generate insights, visualize results, and receive proactive recommendations. The system supports enterprise-grade security, seamless data integration, and customizable knowledge bases tailored to each organization.

Roles and Responsibilities:

- **Developed the FastAPI backend**, which powers the API layer for query processing, document ingestion, and model inference.
- **Integrated Hugging Face Transformers (SentenceTransformers)** to generate high-quality embeddings for text, documents, and structured data used in the retrieval pipeline.
- **Implemented PGVector (PostgreSQL + pgvector extension)** to store, index, and retrieve vector embeddings efficiently as part of the RAG workflow.
- **Built data ingestion pipelines** that allow users to upload or connect their data from various sources such as Google Drive, HubSpot, CSV/Excel files, and cloud storage providers.
- **Designed and implemented the RAG pipeline**, including:
 - Document preprocessing and chunking
 - Embedding generation and vector storage
 - Semantic search for retrieving the most relevant context
 - Passing retrieved context to the LLM for accurate, grounded responses
- **Developed custom knowledge-base chatbots**, enabling organizations to create domain-specific AI assistants using their own documents, CRM data, and business records.
- **Ensured secure processing and data isolation**, implementing authentication, role-based access, and safe handling of sensitive enterprise data.

5.SDTM Issue Tracker (Clinical Trial Data Quality System)

Role: Backend Developer

Description

The SDTM Issue Tracker is a clinical trial data quality management system designed to identify, track, and resolve issues across multiple medical studies. It helps research teams maintain SDTM compliance by managing issue states, recording affected records, and tracking all changes for audit purposes. Users can create issues, assign them to team members, add comments, and move them through a defined workflow: **New** → **In Progress** → **Review** → **Approved** → **Resolved**.

The platform includes dashboards, filters, and analytics to monitor issue status, priorities, and team performance. The backend is built using FastAPI, SQLAlchemy, and MySQL with secure authentication, history logging, and role-based permissions.

Roles & Responsibilities:

- Developed FastAPI-based REST APIs for issue creation, updates, workflow transitions, comments, and assignments.
 - Designed SQLAlchemy models for Issues, Studies, Domains, Comments, History, and User Roles.
 - Implemented workflow automation to manage issue lifecycle states.
 - Added JWT authentication and role-based access control for secure system usage.
 - Built backend logic for dashboards and analytics (priority counts, user assignments, study summaries).
 - Implemented activity timelines and complete audit trails for regulatory compliance.
 - Optimized API performance with efficient queries and async operations.
 - Provided API documentation and supported frontend teams during integration.
 - Managed Docker-based deployment and contributed to backend testing and quality checks.
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6. Solstice Health (AI-Based Project)

Role: Backend Developer

Description: Solstice Health is an AI-driven engine designed to assist life sciences marketing teams in accelerating the creation and evaluation of compliant content.

Roles and Responsibilities:

- Contributed to the development of the system architecture, ensuring proper file structure breakdown for better maintainability.
 - Played a pivotal role in improving backend efficiency by developing robust APIs, optimizing data flow, and enhancing overall system performance.
 - Developed user management features to streamline operations and enhance administrative control.
 - Built APIs for operations such as deleting annotations and managing database records.
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7. SNiP Nutrigenomics

Role: Backend Developer

Description: Developed a DNA-based custom health product application where products are personalized based on customers' DNA reports in an MLM-based business model.

Roles and Responsibilities:

- Enhanced the personalized customer experience by integrating DNA reports for product recommendations.
 - Improved backend functionalities, including White Labeling and Admin Page, increasing administrative efficiency.
 - Developed an Admin Panel that streamlined the management of user commissions and report distribution to Wellness Pro Advisers.
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8. Music Recommendation System

Role: ML Engineer

Description: Developed a music recommendation system that uses Machine Learning (ML) and Natural Language Processing (NLP) techniques for personalized music suggestions.

Roles and Responsibilities:

- Built a recommendation system using CountVectorization and cosine similarity.
- Processed a large dataset (100,000 entries) to optimize the recommendation model.

- Integrated the Saavn API to fetch music details and developed a user-friendly interface with Streamlit.
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8.Cuisine Predictor (Using Logistic Algorithm)

Role: ML Engineer

Description: Developed a Cuisine Predictor system that predicts a user's preferred cuisine using Machine Learning techniques and logistic regression.

Roles and Responsibilities:

- Designed and trained a logistic regression model for cuisine prediction.
 - Built a user interface using Flask and Streamlit.
 - Deployed the system on OnRender for scalability and accessibility.
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9.MOA (MyOutdoor Agent)

Role: Full Stack Developer

Description: Developed and maintained web features for the MOA WebApp, focusing on user experience and functionality.

Roles and Responsibilities:

- Managed the Admin Module, including user sign-in, sign-up, and subscription management.
 - Created stored procedures, optimizing code efficiency and managing application functionality.
 - Developed views to reduce query load and improve performance.
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10.Mill Data (Admin)

Role: Backend Developer

Description: A real-time platform providing insights into mill capacities across North America, offering industry professionals critical decision-making data.

Roles and Responsibilities:

- Developed user pages, including login, password recovery, and client profiles.
- Created stored procedures for efficient CRUD operations.
- Enhanced and maintained web features for improved usability and functionality.
- Managed the Admin Module for user authentication and subscription management.